

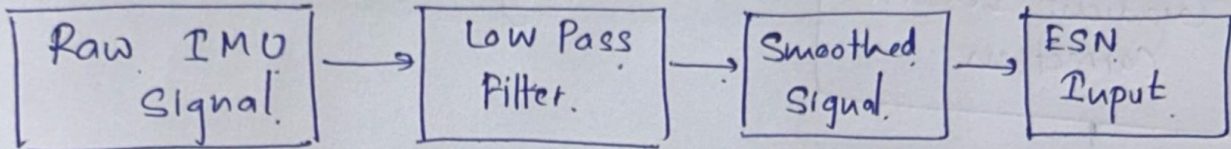
Filter options:

- Low Pass Filter
- Madgwick Filter
- Magnet. Kalman Filter

① LPF

* High freq. removed.

↳ Noise. Assumed High frequency



Basic Equations:

$$y(k) = \alpha x(k) + (1-\alpha)y(k-1)$$

$x(k)$ - current sample.

$y(k)$ - Filtered sample.

α - smoothing factor ($0 < \alpha < 1$)

* Good for basic noise removal.

✓ Simple Implementation

✓ Low Computational cost

✓ Fast Execution

✗ Cannot Estimate Orientation

✗ Cannot Remove Drift

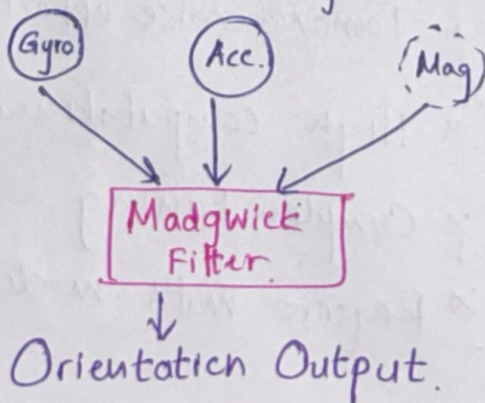
✗ May remove useful vibration info

② Madgwick Filter

Sensors used - Gyroscope.

Accelerometer

Magnetometer (Optional)



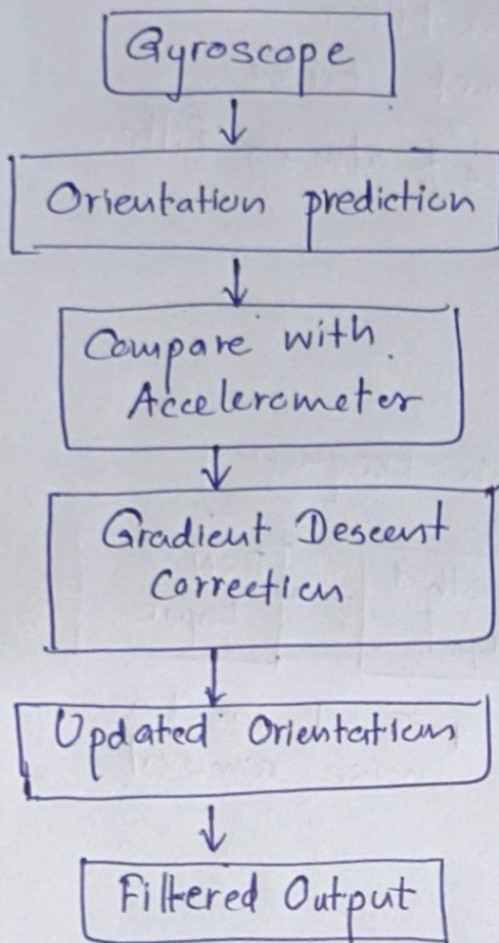
Basic Equations:

$$q(k+1) = \alpha x(k) + \dot{q} \Delta t$$

$x(k)$ - orienting sample

$y(k)$ - filtered output

α - smoothed factor ($0 < \alpha < 1$)



✓ Light Weight

✓ Fast

✓ Sensor Fusion

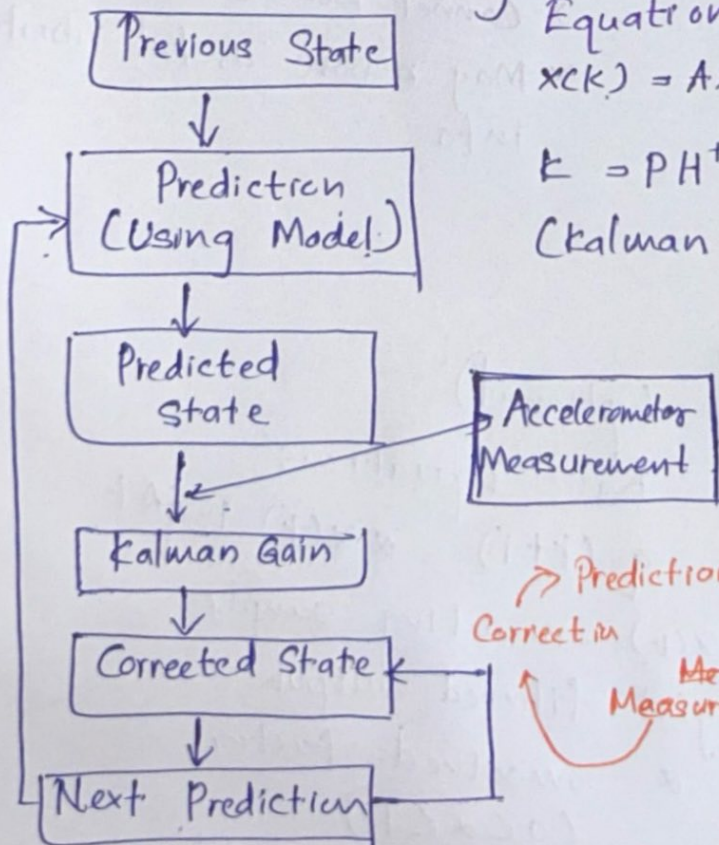
✓ Suits with Arduino, Jetson-Nano

Gyroscope Calibration ?

Accuracy ?

③ Kalman Filter

- Predicts the next state using the math. model
- Corrects it using sensor measurement



Equations:

$$x(k) = Ax(k-1) + Bu(k) \quad (\text{prediction step})$$

$$K = PH^T (HPH^T + R)^{-1}$$

(Kalman Gain)

✓ Excellent Accuracy

✓ Optimal Estimation

✓ Removes noise effectively

✗ High computational cost

✗ Complex tuning

✗ Require math model